NumPy ↔ SQL Conversion Cheat Sheet

SQL: `SELECT column2 FROM table1;`

NumPy: `arr[:, 1]`

SQL: `SELECT column1,column3 FROM table1;`

NumPy: `arr[:, [0,2]]`

SQL: `SELECT * FROM table1 WHERE column1>5;`

NumPy: `arr[arr[:,0]>5]`

SQL: `SELECT column2 FROM table1 WHERE column1>5;`

NumPy: `arr[arr[:,0]>5,1]`

SQL: `SELECT column1+1,column2+1,column3+1 FROM table1;`

NumPy: `arr+1`

SQL: `SELECT PI()*POWER(radius,2) FROM table1;`

NumPy: `np.pi * arr**2`

SQL: `SELECT arr1+arr2 (same row);`

NumPy: `arr1 + arr2`

SQL: `SELECT a.value+b.value FROM arr1 a JOIN arr2 b ON a.id=b.id;`

NumPy: `No direct NumPy join; use structured arrays/boolean indexing (pandas preferred)`

SQL: `SELECT MIN(value) FROM arr1;`

NumPy: `np.min(arr1)`

SQL: `SELECT SUM(column2) FROM arr;`

NumPy: `np.sum(arr[:,1])`

SQL: `SELECT * FROM arr WHERE age>18 AND age<65;`

NumPy: `arr[(arr>18) & (arr<65)]`